

Section 6 patch: corrected leakage constants, the exact band structure, and the $O(y^3)$ extension

Patch note compiled against the June 11 certificate chain

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Summary of changes. (i) The C-even hopping of Theorem 6.2 omits a vacuum-mediated second-order route present for every C-even pair; the corrected per-neighbor hop is $t_+ = -\frac{481}{612} + \frac{3}{4} = -\frac{11}{306}$, and the $k = 0$ coefficient $-\frac{9397}{1020}$ becomes $-\frac{217}{1020}$. The manuscript's own Section 7 domino data adjudicate the correction. (ii) The open signed-band problem of Theorem 6.3 is solved: the lowest C-odd branch is *exactly flat* at $O(y^2)$, with coefficient $\frac{11}{306}$, and remains exactly flat at $O(y^3)$ with cubic coefficient $-\frac{109151}{249696}$. (iii) Remark 6.4's per-bond ratio 5/481 does not survive the correction (the corrected ratio is 5/22); the immobility statement survives in the stronger flat-band form. (iv) The C-odd channel is T_1^{+-} (axial-vector-like), not scalar; the C-even rest-frame content is $A_1^{++} \oplus E^{++}$ with the new E^{++} coefficient $\frac{223}{1020}$.

PATCH 1 — insert after Lemma 6.1:

Lemma 6.1 (Vacuum-mediated route; 6.1'). *For C-even single-flux states the second-order effective Hamiltonian contains the vacuum-intermediate route*

$$\langle e_r | W R W | e_i \rangle \supset \frac{\langle e_r | W | 0 \rangle \langle 0 | W | e_i \rangle}{8/3 - 0} = (\sqrt{2})(\sqrt{2})\frac{3}{8} = +\frac{3}{4} \quad (19')$$

for every ordered pair (i, r) , adjacent or distant, with the same normalization as the operator identity displayed in Theorem 6.2 ($\langle G_p | W_p = (\chi_p + \bar{\chi}_p)(\chi_{p'} + \bar{\chi}_{p'})\langle 0 | \sqrt{2}$). It is absent in the C-odd sector, where $\langle 0 | W | 0 \rangle = 0$ by parity. For a distant pair the route exactly cancels the free two-flux channel sum $2/(8/3 - 16/3) = -\frac{3}{4}$, recovering the vanishing of distant hops; for shared-link neighbors the two-flux sum is the fused-channel value $-\frac{481}{612}$ of Theorem 6.2 and the cancellation is partial.

Proof. The amplitude $\langle e_r | W | 0 \rangle = \sqrt{2}$ per flux state follows from the displayed identity; the intermediate vacuum sits at energy 0 below the single-flux manifold at 8/3, giving the denominator factor 3/8. For a distant pair the only other intermediates are the free two-flux states at 16/3, whose summed squared amplitude is 2; the two routes cancel. The mechanism, both cancellations, and the adjacent-pair value are machine-certified (su3_domino_d3.py, 251 exact-rational gates), and the Section 7 exact diagonalization adjudicates independently: the C-even domino levels $\{1769/3060, 13/20\}$ are $\text{diag} \pm |t_+|$ with $\text{diag} = 1879/3060$, forcing $|t_+| = 110/3060 = 11/306$, not 481/612. $\square$

PATCH 2 — replace Theorem 6.2 (eqs. (20)–(22)) with:

Theorem 6.2 (C-even leakage; corrected). *Per shared-link neighbor, in the C-even sector:*

$$\text{self-energy} = -\frac{481}{612} \quad \left(\text{channel split: } 1 : -\frac{1}{12}, 8 : -\frac{16}{51}, 3 : -\frac{1}{6}, 6 : -\frac{2}{9} \right), \quad (20)$$

independent of the relative link orientation, with vacuum subtraction +3/4 per neighbor on the diagonal. The hopping is the fused-channel sum completed by the vacuum-mediated route of Lemma 6.1':

$$t_+ = -\frac{481}{612} + \frac{3}{4} = -\frac{11}{306}. \quad (20')$$

The operator identity $\langle G_{p'} | W_p = \langle G_p | W_{p'}$ equates the channel sums of hopping and self-energy; the full hop differs from the self-energy by the $|0\rangle$ route, which contributes off-diagonally but is removed from the diagonal by the vacuum subtraction. Assembling at $k = 0$ in the uniform A_1^{++} projection with neighbor count $4(2d - 3)$ in d spatial dimensions,

$$m_+(y) = 4 \Delta_+ \left(\frac{3y}{2} \right) + y^2 L(d) + O(y^3), \quad L(d) = 4(2d - 3) \left(-\frac{11}{153} \right), \quad (21)$$

and in particular, for $d = 3$,

$$m_+(y) = \frac{8}{3} - y - \frac{217}{1020} y^2 + O(y^3), \quad \frac{217}{1020} = 0.2127 \dots \quad (22)$$

[Optional one-line addition to Assumption 6.5, after “the vacuum path contributing +3/4 exactly once”:] “; its off-diagonal counterpart is the route of Lemma 6.1'.”

PATCH 3 — insert after Theorem 6.3:

Theorem 6.3 (Exact $O(y^2)$ band structure; flat C-odd band; 6.3'). *Plaquettes of the cubic lattice carry three orientations, so the one-excitation effective Hamiltonian at $O(y^2)$ is a 3×3 Bloch problem at each k . Let $A(k)$ and $S(k)$ be the unsigned and signed Bloch matrices of the shared-link adjacency (12-regular; signs from the relative orientation $s = \pm 1$ of Theorem 6.3), and let $N(k), \tilde{N}(k)$ be the Bloch incidence matrices of the oriented boundary chains, with $u_j = 1 - e^{ik_j}$, $v_j = 1 + e^{ik_j}$. Then*

$$A(k) + 4I = N(k)N(k)^\dagger, \quad \det N(k) = -2v_1v_2v_3; \quad S(k) + 4I = \tilde{N}(k)\tilde{N}(k)^\dagger, \quad \det \tilde{N}(k) \equiv 0, \quad (23')$$

so $\lambda(k) \in [-4, 12]$, $\mu(k) \in [-4, 8]$, and the C-odd spectrum contains $\mu = -4$ at every k , with kernel vector $w(k) = (\bar{u}_3, -\bar{u}_2, \bar{u}_1)$: the lowest C-odd branch is exactly flat,

$$m_-(k, y) = \frac{8}{3} + y + \frac{11}{306} y^2 + O(y^3) \quad \text{for every } k, \quad (23'')$$

replacing the interval $[-\frac{3}{102}, \frac{17}{102}]$ of Theorem 6.3 by the exact value $\frac{11}{306} = \frac{7}{102} - \frac{20}{612}$. The flat band is spanned by compactly localized elementary-cube 2-chains ($\partial_2 \partial_3 = 0$); on an L^3 torus the $\mu = -4$ eigenspace has dimension $L^3 + 2$. The dispersive C-odd partner branches span $\mu \in [-4, 8]$, i.e. coefficients $[\frac{11}{306}, \frac{41}{306}]$, with $\mu = -4 + |k|^2 + O(k^3)$ (doubly degenerate) near rest. In the C-even sector, with the corrected hop of Theorem 6.2,

$$E_+(k) = \frac{8}{3} - y + y^2 \left[\frac{223}{1020} - \frac{11}{306} \lambda(k) \right], \quad \lambda(k) \in [-4, 12], \quad (23''')$$

with band bottom $-\frac{217}{1020}$ at $k = 0$ ($\lambda = 12$, $\lambda(k) = 12 - \frac{4}{3}|k|^2 + O(k^4)$ isotropically, hence curvature $+\frac{22}{459}|k|^2 y^2$ and effective mass $m^* = 459/(44 y^2)$), band top $\frac{1109}{3060}$ ($\lambda = -4$), and bandwidth $16|t_+| = \frac{88}{153}$. At rest the C-even triplet decomposes as $A_1^{++} \oplus E^{++}$: the A_1^{++} (0^{++}) state is the band bottom, and the E^{++} doublet (2^{++} -like) sits at $\lambda = 0$ with exact, hop-independent coefficient $\frac{223}{1020}$. The C-odd rest triplet is T_1^{+-} (1^{+-} -like): the C-odd one-plaquette sector contains no scalar at rest.

PATCH 4 — insert after Theorem 6.3':

Theorem 6.4 (Third order: tromino vanishing and the rigid flat band; 6.3''). At $O(y^3)$ the only new lattice geometry is the tromino (three plaquettes pairwise sharing links). Every $O(y^3)$ tromino weight vanishes (bare-link lemma): each candidate three-plaquette process leaves at least one boundary link bare, and the corresponding Haar integral vanishes; trominoes first activate at $O(y^4)$. Consequently the third-order effective Hamiltonian retains the second-order signed-adjacency structure, with C-odd hopping odd in s ($T_3(s) = -T_3(-s)$), and the flat band of Theorem 6.3' is rigid at third order:

$$m_-(k, y) = \frac{8}{3} + y + \frac{11}{306} y^2 - \frac{109151}{249696} y^3 + O(y^4) \quad \text{for every } k, \quad (23''''')$$

with $d_3 = \frac{7}{32} + 12 \text{leak}_3 - 4 b_3 = -\frac{109151}{249696} \approx -0.437135$, where $b_3 = \frac{1975}{124848}$ is the third-order C-odd hop and $\text{leak}_3 = -\frac{12331}{249696}$ the per-neighbor diagonal leakage; the C-odd dispersive top has cubic coefficient $-\frac{61751}{249696}$. In the C-even sector at rest,

$$m_+(0, y) = \frac{8}{3} - y - \frac{217}{1020} y^2 - \frac{54049}{520200} y^3 + O(y^4), \quad (23''''')$$

and at the band top ($\lambda = -4$) the cubic coefficient is $\frac{471353}{1560600}$. The diagonal and hopping corrections obey the exact identity $\text{leak}_n^e = T_n^e$ at $n = 2, 3$ (two distinct $+3/4$ mechanisms of equal value), so the A_1^{++} correction at $k = 0$ is (within-tower) $+24 T_n^e$ per order.

PATCH 5 — replace Remark 6.4 with:

Remark 6.5 (An exactly immobile excitation; corrected). With the corrected hop the per-bond ratio is $|t_-|/|t_+| = (5/612)/(11/306) = 5/22$, so the ratio no longer carries an immobility claim. The claim holds in a stronger, exact form (Theorems 6.3'–6.3''): the lowest C-odd branch is exactly dispersionless through $O(y^3)$ — bandwidth 0, infinite effective mass at these orders — while the full C-odd three-branch manifold has width $\frac{5}{51}$ in y^2 units against the C-even bandwidth $\frac{88}{153}$ (manifold-width ratio $(5/51)/(88/153) = 15/88$). The flatness is structural ($\partial_2 \partial_3 = 0$ on the signed incidence complex), not a small-parameter accident. This is, to our knowledge, a new structural observation about the strong-coupling T_1^{+-} sector.

PATCH 7 — append to the Appendix A certificate list:

- `su3_moments_ext.py` (27 gates) — extended exact SU(3) Haar moment engine: Wein-garten $p = q \leq 3$, charge- ± 3 ε -projector blocks, independent Weyl-torus oracle.
- `su3_domino_d3.py` (251 gates) — abstract-domino des Cloizeaux calculus to $O(y^3)$: the vacuum-route lemma, $t_+ = -11/306$, the Section 7 levels as exact gates, b_3 , leak_3 , $d_3 = -109151/249696$, and the C-even cubic constants.

- `glueball_band_certificate_v2.py` (36 gates) — the $O(y^2)$ band structure of Theorem 6.3': incidence factorizations, $\det \tilde{N} \equiv 0$, cube 2-chains and the $L^3 + 2$ multiplicity, corrected C-even block, O_h quantum numbers; the as-written $-9397/1020$ is retained as a provenance gate.
- `tromino_contract_independent_check.py` (19 gates) and `tromino_candidate_closed_form_check.py` — the $O(y^3)$ tromino-vanishing sector of Theorem 6.3''.

Constant provenance table

Every constant in the patches, with its certificate:

constant	value	certificate
t_+	$-11/306$	<code>su3_domino_d3.py</code> ; §7 adjudication
A_1^{++} at $k = 0$, $O(y^2)$	$-217/1020$	<code>d3_results.json</code> ; band v2
band top, $O(y^2)$	$1109/3060$	<code>d3_results.json</code> ; band v2
E^{++} at $k = 0$	$223/1020$	band v1 and v2 (t -independent)
curvature	$22/459$	band v2; Bloch derivation (master §3.15)
bandwidth	$88/153$	band v2
flat band, $O(y^2)$	$11/306$	band v1/v2; <code>su3_domino_d3.py</code>
b_3	$1975/124848$	<code>su3_domino_d3.py</code>
leak ₃ (C-odd)	$-12331/249696$	<code>su3_domino_d3.py</code>
d_3	$-109151/249696$	<code>su3_domino_d3.py</code>
C-odd dispersive top, $O(y^3)$	$-61751/249696$	<code>su3_domino_d3.py</code>
$m_+(0)$ cubic	$-54049/520200$	<code>su3_domino_d3.py</code>
C-even band-top cubic	$471353/1560600$	<code>su3_domino_d3.py</code>
per-bond ratio	$5/22$	band v2
manifold-width ratio	$15/88$	band v2